MICRO 2026
Plastic Pollution from Macro to nano
Cádiz, 12–16 October 2026

Walking Talk · Group 10
13 October, 12:30–14:00

Screening commercial smoked seafood for microplastics: a multi-taxa market survey

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Why smoked seafood?

- **Eaten as sold.** Smoked seafood is ready to eat and often eaten whole or near-whole. What the pack carries, the consumer eats.
- **A gap in the data.** Processed seafood is under-represented in microplastic surveys compared with fresh fish and shellfish.
- **More contact surfaces.** Brining, smoking racks, slicing and vacuum film add contact points that fresh-product surveys never sample.
- **Aim.** A transparent, reproducible baseline for ready-to-eat seafood on the Turkish market, as the basis for dietary-exposure assessment.

Sold sealed, ready to eat › Eaten as sold, often whole › What else is in the pack?

What we sampled

45 packs · 15 categories · every category in triplicate

Packs were bought between September 2025 and April 2026 from chain retailers and specialist fishmongers in Istanbul, carried in a cool box and kept sealed at −20 °C; species, brand, producer, lot and pack weight were recorded. Atlantic salmon comes from five anonymised suppliers (A to E) plus supplier A's ASC-certified line, so sourcing and certification can be compared within one species.

Atlantic salmon 6 categories · **Other finfish** 7 categories · **Crustacean** 1 category · **Cephalopod** 1 category

Salmon A *Salmo salar* 100 g pack	**Salmon B** *Salmo salar* 200 g pack	**Salmon C** *Salmo salar* 100 g pack	**Salmon D** *Salmo salar* 100 g pack	**Salmon E** *Salmo salar* 100 g pack
Salmon A ASC *Salmo salar* 100 g pack	**Rainbow trout** *Oncorhynchus mykiss* 100 g pack	**Black Sea trout** *Salmo labrax* 100 g pack	**Mackerel** *Scomber scombrus* 200 g pack	**Mackerel (çiroz)** *Scomber scombrus* 1000 g pack
Tuna (pouch) *Thunnus alalunga* 80 g pack	**Sea bream** *Sparus aurata* 100 g pack	**Sea bass** *Dicentrarchus labrax* 100 g pack	**Shrimp** *Penaeus spp.* 100 g pack	**Octopus** *Octopus vulgaris* 250 g pack

How we looked

10 g from each pack per digest, contamination-controlled, pictured at our bench

1 Open and weigh

Thawed overnight at +4 °C. Pack interior rinsed three times with filtered ultrapure water, rinse pooled with the sample. A 10 g portion weighed to 0.01 g per digest.



A pack as bought: tray, film and fillets.

2 Alkaline digestion

10 % KOH with 2 % Tween-20, 1:5 (w/v), 40 °C, 150 rpm, 24 h in foil-capped glass flasks. The surfactant keeps the fat emulsified so the filter does not clog.



Foil-capped flasks in the shaking incubator.

3 Neutralise

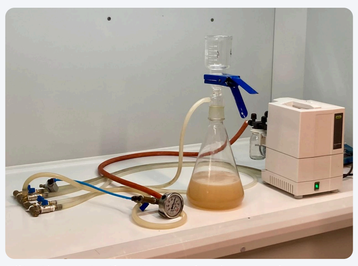
Diluted 1:1 with filtered water, then 1 M citric acid under stirring to pH 7.5, the stability window of the alumina membrane.



Digests on the multi-stirrer, pH meter alongside.

4 Filter and clean

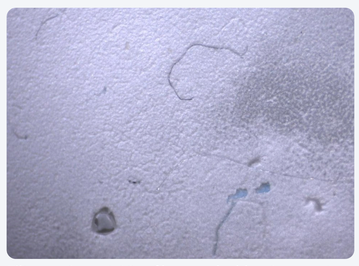
0.2 µm Al_2O_3 membrane, 47 mm, at −500 mmHg; funnel rinsed 3 × 10 mL. 15 % H_2O_2 on the filter for 30 min, then dried 24 h in a covered dish.



Filtration onto the 47 mm alumina membrane.

5 Count and classify

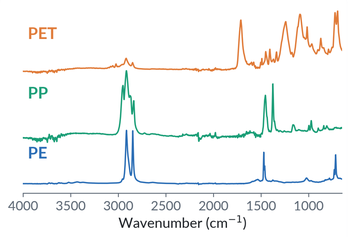
Stereo microscope: whole filter at 10×, each particle at 40×. Morphotype, colour and maximum length (ImageJ) recorded per particle.



Filter surface at 10×: fibres and a blue fragment.

6 Next: polymer identity

µ-FTIR and ATR on 3–5 particles per filter, stratified by size; then exposure per 100 g and 225 g portion in the thesis.



Reference ATR-FTIR spectra from our polymer library.

45 retail packs, 15 smoked products in triplicate

100 % of packs with suspected particles (blanks pending)

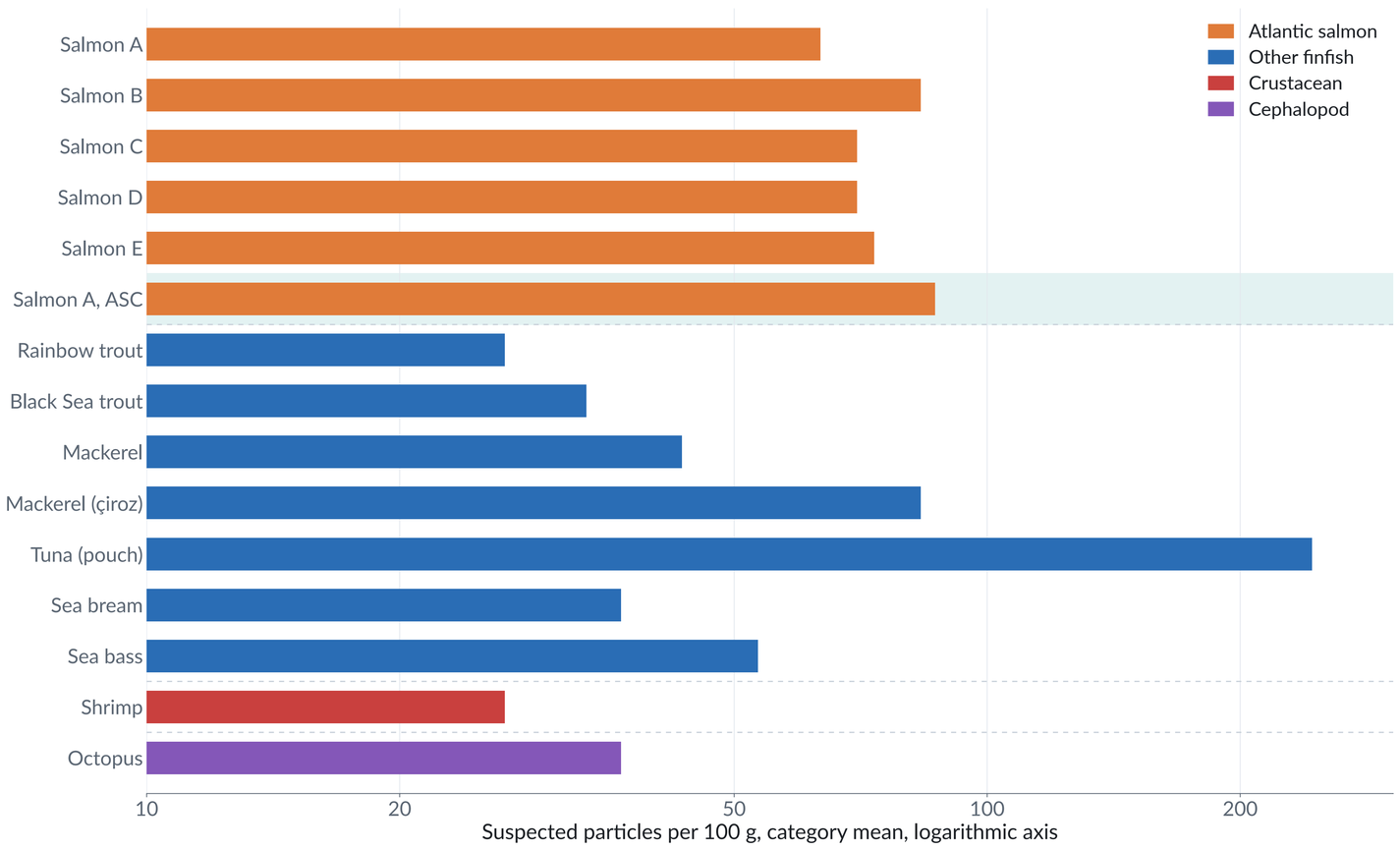
50 particles per 100 g, median

81 % Fibres, the dominant morphotype

72 % of particles 1–5 mm long

What we found

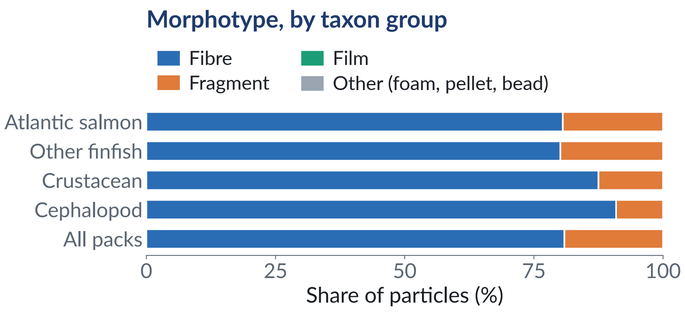
suspected particles per 100 g, by product category · preliminary

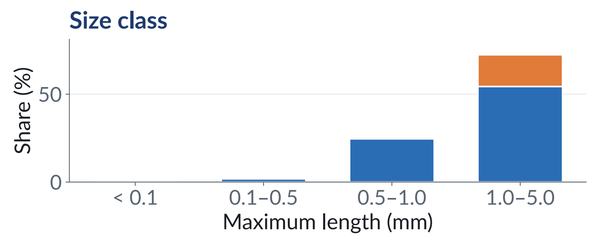


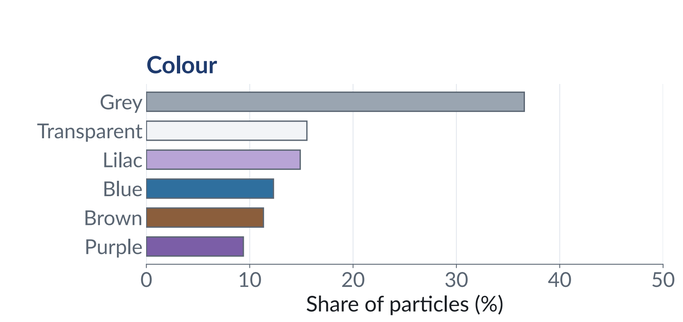
Bars: mean of the three packs of a category (raw counts, blank subtraction pending), on a logarithmic axis. Shaded row: the ASC-certified line, from the same supplier as Salmon A.

What they look like

309 particles on 45 filters, uncorrected







Take-home

1. **Occurrence.** Suspected microplastics were recovered from 100 % of the 45 packs analysed, in every taxon group; fibres made up 81 % of all particles, 72 % were 1–5 mm long and grey was the most frequent colour (37 %).
2. **Within Atlantic salmon.** Supplier means ranged from 63 to 87 particles per 100 g across suppliers A to E; the ASC-certified line (87) was not lower than the same supplier's standard line (63).
3. **Preliminary, and an upper bound.** First-pass counts; blank correction is pending. Visual identification over-counts: polymer identity by µ-FTIR comes next, and only confirmed particles enter the per-portion exposure estimate (100 g and 225 g) in the thesis.

Talk to me

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Find me: Walking Talk group 10, 13 to 15 October, 12:30–14:00; this talk is on 13 October. Write to me with questions or your own retail-pack numbers.

1. Dehaut A, et al. Microplastics in seafood: benchmark protocol for their extraction and characterization. Environ Pollut 2016;215:223–33.
2. Hove HTB, et al. Quick and efficient microplastic isolation from fatty fish tissues by surfactant-enhanced alkaline digestion. Mar Pollut Bull 2023;197:115726.
3. Duman S, et al. Optimization of a method designed to extract and characterize microplastics in different packaged fish products. Food Control 2023;154:110029.
4. Crawford CB, Quinn B. Microplastic Pollutants. Elsevier; 2016.

Funding. This study was funded by Scientific Research Projects Coordination Unit of Istanbul University. Project number: FYL-2025-41504. **Conflict of interest.** The authors have no conflict of interest to disclose related to this presentation.